

5.3.1 Secure Collaboration (Authentication, MFA, Access Control)

محفوظ تعاون کا مطلب ہے ڈیجیٹل ٹولز سے محفوظ، کنٹرول شدہ اور ذمہ دارانہ انداز میں ساتھ کام کرنا۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 09 of 17

★ Golden

5.3.1 Secure Collaboration (Authentication, MFA, Access Control)

★ Golden

LEARN IT

Digital tools allow people to **share files, attend online classes, edit documents and work together from different locations**. Collaboration is useful only when **data is protected** and **communication remains safe**.

Secure collaboration means using digital tools to work together in a **safe, controlled and responsible** way. It depends on several key elements.

i) Authentication / identity verification

Authentication is **confirming the identity of a user before allowing access**. Common methods:

- **Username and password**
- **PIN** (a short number code)
- **OTP — One-Time Password**, a code that works only once, usually sent by SMS or an authenticator app
- **Biometrics** — fingerprint or face

Source example: a **banking app sending an OTP to your phone before login**.

Multi-factor authentication (MFA) and two-factor authentication (2FA)

A **factor** is a category of proof:

- **Something you know** — password or PIN.
- **Something you have** — your phone, an OTP, a bank SMS.
- **Something you are** — fingerprint, face (biometrics).

Multi-factor authentication (MFA) means the system asks for **two or more different factors** before it trusts you. **Two-factor authentication (2FA)** is the common case of MFA that uses **exactly two** factors — usually password + OTP.

If a thief in another city steals only your password, MFA still blocks them because they do not have your phone (or your fingerprint).

ii) Access control

Access control decides **what an authenticated user is allowed to do**: View, Edit, Comment, Download, Share.

Authentication answers **"Who are you?"** Access control answers **"What may you do?"**

Source example: a teacher shares a Google Doc with students as **"view only"**, but gives group leaders **"edit"** permissions.

iii) Accountability and tracking

User actions are linked to **verified accounts**. You can see who opened, edited, deleted or shared a file. Example: a school **LMS** (Learning Management System) tracks which teacher uploaded an assignment and which student submitted it.

iv) Data privacy

Data privacy means protecting personal and sensitive information and using it **only for the purpose it was collected**. It involves:

- **Protection** — keeping names, numbers, marks and passwords secure.
- **Controlled sharing** — sharing only with the right people.
- **Confidentiality** — data stays private during storage and transmission.
- **Compliance** — following legal and ethical rules for data handling.

v) Safe communication / transmission

Exchanging information through digital platforms **securely**, protecting messages from unauthorised interception or wiretapping. Example: a teacher shares class instructions through an **official, encrypted LMS** rather than an open public forum or an unknown WhatsApp broadcast.

WORKED EXAMPLE

Worked example: a Class XII Google Classroom in Sukkur.

Step 1 — Authentication. Each student signs in with a school email and password. The college turns on MFA: after the password, Google sends an OTP to the student's registered Jazz/Zong number.

Step 2 — Access control. All 40 students can **view** the assignment sheet. Only the four group leaders can **edit** the shared report. Nobody except the teacher can **delete** the class materials. A student who left the college is removed from the class (access revoked).

Step 3 — Accountability. Version history shows that Ali edited paragraph 2 at 8:14 pm and that Hina submitted the PDF at 9:02 pm.

Step 4 — Privacy and safe transmission. Marks stay inside Classroom; the teacher does not post CNIC numbers or phone numbers in a public Facebook group. Traffic to Google is encrypted, unlike an open public forum.

If Ali shares his password with a cousin, authentication is broken. If the teacher gives every student "edit" on the marksheet, access control is broken even though login still works.

REMEMBER IT

A-A-A-P-S. Authentication, Access control, Accountability, Privacy, Safe transmission. MFA = at least two different factor types (know / have / are). 2FA is MFA with two factors.

COMMON MISTAKE

Mistake: Treating authentication and access control as the same thing.

Fix: Login only proves identity. After login, access control still limits View / Edit / Comment / Download / Share. A student who is authenticated must not automatically get teacher rights.

محفوظ تعاون کا مطلب ہے ڈیجیٹل ٹولز سے محفوظ، کنٹرول شدہ اور ذمہ دارانہ انداز میں ساتھ کام کرنا۔ تصدیق (Authentication) یہ جانچتی ہے کہ صارف کون ہے؛ رسائی کنٹرول (Access Control) یہ طے کرتا ہے کہ وہ کیا کر سکتا ہے (صرف دیکھنا، ترمیم، تبصرہ)۔ کثیر عنصری تصدیق (MFA) دو یا زیادہ مختلف ثبوت مانگتی ہے، جیسے پاس ورڈ اور او ٹی پی (OTP)۔ حساب دہی، ڈیٹا پرائیویسی اور محفوظ ترسیل بھی ضروری ہیں۔

EXAM TIP

Typical items: "Define secure collaboration", "Differentiate authentication and access control", "What is MFA/2FA?", "Explain five elements of secure collaboration with examples."

Write the five headings (authentication, access control, accountability, privacy, safe transmission). For MFA, name the three factor types and say that 2FA is password + OTP in a banking app. Use the Google Doc "view only vs edit" example — examiners look for it.

CHECK YOURSELF

1. What question does authentication answer, and what question does access control answer?
2. Password + OTP is an example of which idea?
3. Give the source example of access control in a Google Doc.

Answers: 1. Authentication: who are you? Access control: what may you do? 2. Two-factor / multi-factor authentication (2FA/MFA). 3. Teacher shares the document as view-only with students but gives group leaders edit permission.

Next lecture: 5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)